



PRESIDENCY UNIVERSITY

Itgalpura, Rajanukunte, Bengaluru - 560064

School of Engineering/School of Computer Science and Engineering

A Project Report on

“UV Exposure Monitor and Warning Wearable”

Submitted in partial fulfillment of the requirements for the course
Design Workshop (ECE 1511)

Submitted by

Group No- :INV 104

Student Name	Roll No
SRISHANT	20251CSE0743
ANANYA VALASANG	20251CSE0739
DEEKSHITHA K N	20251CSE0745
PUNITH K G	20251CSE0760
K NIKHIL	20251CSE0721

Under the supervision of

Dr.MANIKANDAN,Professor
Department Of Electronics And Communication Engineering
Department Of ECE

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This project presents a wearable UV exposure monitoring device designed to enhance personal safety by providing real-time environmental data. The system is built using an ESP32-C3 microcontroller interfaced with multiple sensors, including a UV sensor (LTR390), temperature and pressure sensor (BMP280), and ambient light sensor (BH1750), all connected via I2C communication. The collected data is displayed on an OLED screen in an intuitive dashboard format. A buzzer is integrated to alert users when UV levels exceed safe thresholds, helping prevent prolonged exposure.

The device is powered by a rechargeable lithium-ion battery with a TP4056 charging module and includes a switch for power control, making it portable and energy-efficient. This solution demonstrates a practical approach to wearable health monitoring and has potential applications in outdoor safety, preventive healthcare, and IoT-based environmental sensing systems.

INTRODUCTION:

Background:

With increasing awareness of environmental health, monitoring exposure to ultraviolet (UV) radiation has become an important concern. Prolonged exposure to UV radiation can lead to serious health issues such as skin damage, premature aging, and an increased risk of skin cancer. In countries with high sunlight intensity, especially in tropical regions, individuals are often exposed to harmful UV levels without real-time awareness. Existing solutions, such as mobile weather applications, provide generalized UV index data but fail to offer personalized, real-time exposure monitoring.

Problem:

The main problem lies in the lack of a portable, affordable, and real-time system that continuously tracks UV exposure and alerts users instantly. People engaged in outdoor activities, such as construction workers, athletes, and travelers, are particularly vulnerable due to prolonged sun exposure without proper monitoring tools.

Solution for the problem:

To address this issue, this project proposes a wearable UV exposure monitoring device based on the ESP32-C3 microcontroller. The system integrates multiple sensors to measure UV radiation, ambient light, temperature, and pressure, providing a comprehensive environmental overview. The data is displayed on an OLED screen, and a buzzer alerts the user when UV levels exceed safe limits. Powered by a rechargeable battery, the device is compact, portable, and suitable for real-world applications. This solution offers a practical and cost-effective approach to preventive healthcare and environmental monitoring.

MOTIVATION:

With increasing levels of ultraviolet (UV) radiation and growing awareness of its harmful effects, there is a strong need for real-time personal exposure monitoring systems. Prolonged exposure to UV radiation can lead to serious health issues such as skin damage, premature aging, and an increased risk of skin cancer. Existing solutions, such as weather applications, provide only generalized UV index data and do not offer continuous, personalized monitoring. This limitation creates a gap for a portable and user-friendly device that can actively track environmental conditions and alert users instantly. Therefore, this project was chosen to develop a wearable UV monitoring system that promotes preventive healthcare, enhances user awareness, and provides a practical solution for individuals exposed to outdoor environments for extended periods.

Objectives:

- * To design and develop a wearable UV exposure monitoring device.
- * To use the ESP32-C3 microcontroller for real-time data processing and control.
- * To measure UV radiation levels accurately using a UV sensor.
- * To integrate additional sensors for temperature, pressure, and ambient light monitoring.
- * To display real-time environmental data on an OLED screen.
- * To implement a buzzer-based alert system for high UV exposure levels.
- * To develop a battery-powered system with a charging module and power switch.
- * To ensure portability, efficiency, and ease of use.
- * To create a cost-effective and scalable solution for IoT-based applications.
- * To promote awareness and preventive healthcare through real-time monitoring.

HARDWARE AND SOFTWARE TOOLS USED:

Hardware tools:

- * ESP32-C3 Microcontroller
- * LTR390 UV Sensor
- * BMP280 (Temperature & Pressure Sensor)
- * BH1750 (Ambient Light Sensor)
- * OLED Display (SSD1306, 0.96 inch)
- * Buzzer (Active)
- * Lithium-ion Battery (3.7V, ~300mAh)
- * TP4056 Charging Module
- * Switch (for power control)
- * Breadboard
- * Jumper Wires

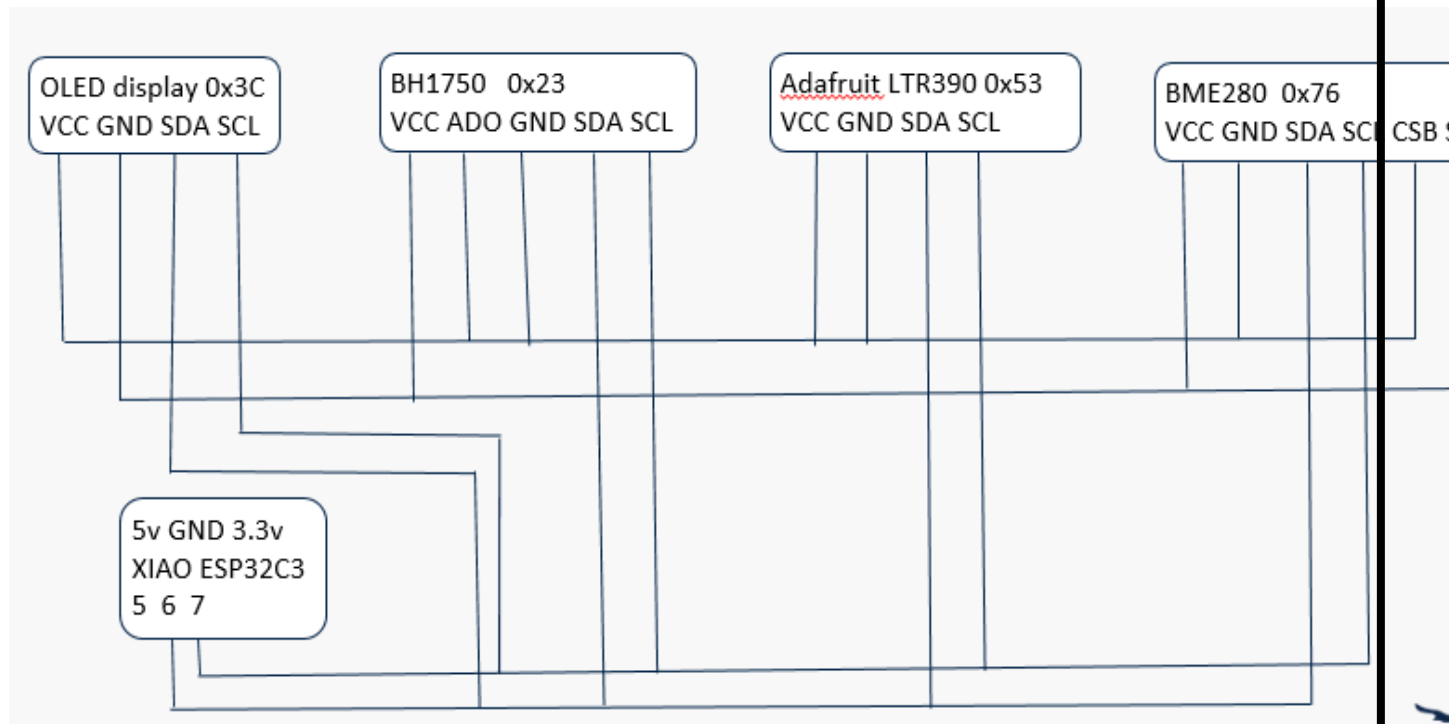
Software tools:

- * Arduino IDE
- * Embedded C / Arduino Programming Language
- * ESP32 Board Package (in Arduino IDE)

Required Libraries:

- * Wire Library (I2C communication)
- * Adafruit LTR390 Library

BLOCK DIAGRAM AND DESCRIPTION:



Description :

The system is a wearable UV monitoring device based on the ESP32-C3 microcontroller. It uses I2C to connect sensors: LTR390 (UV), BMP280 (temperature & pressure), and BH1750 (light). The ESP32 collects and processes environmental data in real time. Data is displayed on an OLED screen in a clear dashboard format. A buzzer alerts the user when UV levels exceed safe limits. The device is compact, portable, and energy-efficient. It also supports future upgrades like wireless connectivity and data logging.

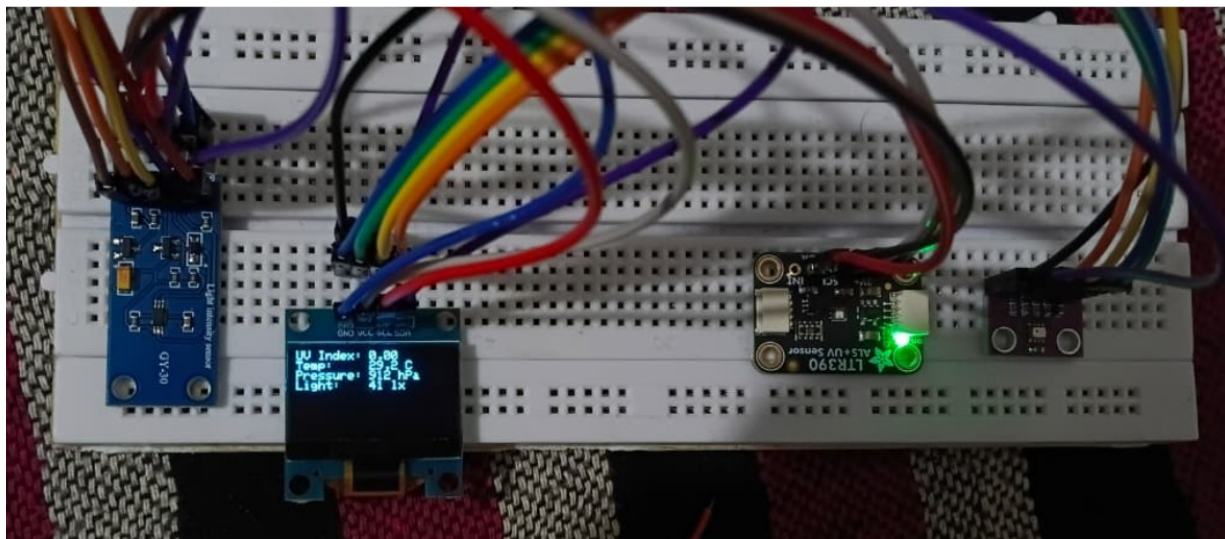
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RESULTS :

- * The UV exposure monitoring device was successfully developed and tested.
 - * It accurately measured UV index, temperature, pressure, and light intensity in real time.
 - * The OLED display showed data clearly in a structured format.
 - * The buzzer provided alerts when UV levels exceeded safe limits.
 - * This feature improved user safety and awareness.
 - * Multiple sensors enabled a more comprehensive environmental analysis.
 - * The device operated reliably with consistent performance.
 - * The battery-powered design ensured portability and ease of use.
 - * It functioned without the need for continuous external power.
- Overall, the system proved efficient, practical, and suitable for real-world use.



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CHALLENGES FACED :

1. Ensuring stable I2C communication between multiple sensors on shared lines
2. Data errors due to loose connections or improper wiring
3. Power instability caused by battery voltage fluctuations
4. Need for proper voltage regulation for consistent performance
5. Time-consuming hardware debugging on breadboard setups
6. Issues like OLED not working or sensors not responding
7. Careful selection of suitable GPIO pins on ESP32-C3

Resolved through systematic testing and proper wiring practices

CONCLUSION :

The project successfully demonstrates the design and implementation of a wearable UV exposure monitoring device using the ESP32-C3 microcontroller. By integrating multiple sensors, the system provides real-time environmental data and alerts users to harmful UV exposure levels. The use of an OLED display and buzzer enhances user interaction and safety by presenting information clearly and providing timely warnings.

The device is compact, portable, and energy-efficient, making it suitable for real-world applications, particularly for individuals exposed to outdoor environments. The project highlights the potential of IoT-based wearable technology in promoting preventive healthcare and environmental awareness. With further enhancements such as wireless connectivity and data analytics, the system can be expanded into a more advanced and scalable solution.

ACKNOWLEDGEMENTS :

We express our sincere gratitude to our guide Dr. M Manikandan for his valuable guidance and support. We also thank the Department of Electronics and Communication Engineering for providing the necessary resources.

PARTICIPATION DETAILS :

Participated in Design Workshop Project Review (Phase -1) .
Developed prototype for UV Index Monitoring and Warning Wearable.